

Theory of Computation (TOC)

Module 1		
S.No.	Concepts	Link
1.	TOC introduction TOC Lec-1 Bhanu Priya	https://www.youtube.com/watch?v=dCiZZiqVv9w
2.	symbol, alphabet, string basic definations TOC Lec-2 Bhanu Priya	https://www.youtube.com/watch?v=jQRUKXhwHa4
3.	sempty, lenght, concatenation of string TOC Lec-3 Bhanu Priya	https://www.youtube.com/watch?v=a0Obv_cJLpl
4.	power of alphabet TOC Lec-4 Bhanu Priya	https://www.youtube.com/watch?v=fnaCgVn29fU
5.	languages & operations TOC Lec-5 Bhanu Priya	https://www.youtube.com/watch?v=dCiZZiqVv9w&list=PLrjkTql3jnmTWSXXvWX1jX-L6f1QJSx
6.	finite automata TOC Lec-6 Bhanu Priya	https://www.youtube.com/watch?v=8g1AUtdayt0&list=PLrjkTql3jnmTWSXXvWX1jX-L6f1QJSx&index=6
7.	transition diagrams, table and function finite automata TOC Lec-7 Bhanu Priya	https://www.youtube.com/watch?v=oaD7zcjGU28&list=PLrjkTql3jnmTWSXXvWX1jX-L6f1QJSx&index=7
8.	finite automata applications & different type TOC Lec-8 Bhanu Priya	https://www.youtube.com/watch?v=2S4At1yakug&list=PLrjkTql3jnmTWSXXvWX1jX-L6f1QJSx&index=8
9.	deterministic finite automata (DFA) TOC Lec-9 Bhanu Priya	https://www.youtube.com/watch?v=OqknGBJqhc&list=PLrjkTql3jnmTWSXXvWX1jX-L6f1QJSx&index=9
10.	dfa example with solution Part-1 TOC Lec-10 Bhanu Priya	https://www.youtube.com/watch?v=O6bwSnoTniY&list=PLrjkTql3jnmTWSXXvWX1jX-L6f1QJSx&index=10
11.	dfa example with solution Part-2 TOC Lec-11 Bhanu Priya	https://www.youtube.com/watch?v=F9FZYPT6X8Q&list=PLrjkTql3jnmTWSXXvWX1jX-L6f1QJSx&index=11
12.	dfa example with solution Part-3 TOC Lec-12 Bhanu Priya	https://www.youtube.com/watch?v=cwWkw8gNCcA&list=PLrjkTql3jnmTWSXXvWX1jX-L6f1QJSx&index=12
13.	non deterministic finite automata (NFA) TOC Lec-13 Bhanu Priya	https://www.youtube.com/watch?v=rKCAPVaU0Qk&list=PLrjkTql3jnmTWSXXvWX1jX-L6f1QJSx&index=13
14.	NFA example Part-1 TOC Lec-14 Bhanu Priya	https://www.youtube.com/watch?v=764ITgEu-KU&list=PLrjkTql3jnmTWSXXvWX1jX-L6f1QJSx&index=14
15.	NFA example Part-2 TOC Lec-15 Bhanu Priya	https://www.youtube.com/watch?v=IcyDv1bWR1k&list=PLrjkTql3jnmTWSXXvWX1jX-L6f1QJSx&index=15
16.	NFA example Part-3 TOC Lec-	https://www.youtube.com/watch?v=Ws3014s6ncY&list=PLrjkTql3jnmTWSXXvWX1jX-L6f1QJSx&index=16

	16 Bhanu Priya	m_TWSXXvWX1_jX-L6f1QJSx&index=16
17.	nfa with epsilon TOC Lec-17 Bhanu Priya	https://www.youtube.com/watch?v=PLrjKtQl3jnm_TWSXXvWX1_jX-L6f1QJSx&index=17
18.	nfa and dfa differences TOC Lec-18 Bhanu Priya	https://www.youtube.com/watch?v=FZp7CJwe6nw&list=PLrjKtQl3jnm_TWSXXvWX1_jX-L6f1QJSx&index=18
19.	conversion of nfa to dfa examples Part-1 TOC Lec-19 Bhanu Priya	https://www.youtube.com/watch?v=syJutOdLnI&list=PLrjKtQl3jnm_TWSXXvWX1_jX-L6f1QJSx&index=19
20.	conversion of NFA to DFA example Part-2 TOC Lec-20 Bhanu Priya	https://www.youtube.com/watch?v=hIIYyVCPKQk&list=PLrjKtQl3jnm_TWSXXvWX1_jX-L6f1QJSx&index=20
21.	equivalence of two finite automata TOC Lec-21 Bhanu Priya	https://www.youtube.com/watch?v=JQTcNUPJlws&list=PLrjKtQl3jnm_TWSXXvWX1_jX-L6f1QJSx&index=21
22.	equivalence of two FA example TOC Lec-22 Bhanu Priya	https://www.youtube.com/watch?v=wtWR6xF-Mk&list=PLrjKtQl3jnm_TWSXXvWX1_jX-L6f1QJSx&index=22
23.	minimization of DFA TOC Lec-23 Bhanu Priya	https://www.youtube.com/watch?v=VeFvQoRuXk&list=PLrjKtQl3jnm_TWSXXvWX1_jX-L6f1QJSx&index=23
24.	minimization of DFA example TOC Lec-24 Bhanu Priya	https://www.youtube.com/watch?v=53qT4NZ3nOo&list=PLrjKtQl3jnm_TWSXXvWX1_jX-L6f1QJSx&index=24
25.	Mealy machine TOC Lec-25 Bhanu Priya	https://www.youtube.com/watch?v=mUtn8il7b4U&list=PLrjKtQl3jnm_TWSXXvWX1_jX-L6f1QJSx&index=25
26.	mealy machine design Example1 TOC Lec-26 Bhanu Priya	https://www.youtube.com/watch?v=Et7Kv-p-acs&list=PLrjKtQl3jnm_TWSXXvWX1_jX-L6f1QJSx&index=26
27.	mealy machine design Example2 TOC Lec-27 Bhanu Priya	https://www.youtube.com/watch?v=Xj-KwpMqGHI&list=PLrjKtQl3jnm_TWSXXvWX1_jX-L6f1QJSx&index=27
28.	mealy machine design Example3 TOC Lec-28 Bhanu Priya	https://www.youtube.com/watch?v=y_G0Xjm5CfI&list=PLrjKtQl3jnm_TWSXXvWX1_jX-L6f1QJSx&index=28
29.	moore machine TOC Lec-29 Bhanu Priya	https://www.youtube.com/watch?v=8mbPgbGXSvs&list=PLrjKtQl3jnm_TWSXXvWX1_jX-L6f1QJSx&index=29
30.	moore machine design Example1 TOC Lec-30 Bhanu Priya	https://www.youtube.com/watch?v=KSjEKhUY9ok&list=PLrjKtQl3jnm_TWSXXvWX1_jX-L6f1QJSx&index=30
31.	moore machine design Example2 TOC Lec-31 Bhanu Priya	https://www.youtube.com/watch?v=xMR2zek_e2c&list=PLrjKtQl3jnm_TWSXXvWX1_jX-L6f1QJSx&index=31
32.	moore machine design Example3	https://www.youtube.com/watch?v=7zj8krqoBw0&list=PLrjKtQl3jnm_TWSXXvWX1_jX-L6f1QJSx&index=32

	TOC Lec-32 Bhanu Priya	m TWSXXvWX1 jX-L6f1QJSx&index=32
33.	mealy and moore machine differences TOC Lec-33 Bhanu Priya	https://www.youtube.com/watch?v=IH_qfCTh9Co&list=PLrjKTqI3jnm TWSXXvWX1 jX-L6f1QJSx&index=33
34.	mealy to moore conversion TOC Lec-34 Bhanu Priya	https://www.youtube.com/watch?v=O1UaAluDgqY&list=PLrjKTqI3jnm TWSXXvWX1 jX-L6f1QJSx&index=34
35.	mealy to moore example TOC Lec-35 Bhanu Priya	https://www.youtube.com/watch?v=dYcMNXjDVVE&list=PLrjKTqI3jnm TWSXXvWX1 jX-L6f1QJSx&index=35
36.	mealy to moore example 2 TOC Lec-36 Bhanu Priya	https://www.youtube.com/watch?v=dKOGsHPelc0&list=PLrjKTqI3jnm TWSXXvWX1 jX-L6f1QJSx&index=36
37.	moore to mealy machine conversion Example-1 TOC Lec-37 Bhanu Priya	https://www.youtube.com/watch?v=DOqa-nI2lGI&list=PLrjKTqI3jnm TWSXXvWX1 jX-L6f1QJSx&index=37
38.	moore to mealy machine conversion Example-2 TOC Lec-38 Bhanu Priya	https://www.youtube.com/watch?v=OqZyBVR9Dr0&list=PLrjKTqI3jnm TWSXXvWX1 jX-L6f1QJSx&index=38
39.	moore to mealy machine conversion Example-3 TOC Lec-39 Bhanu Priya	https://www.youtube.com/watch?v=vm1FQIeQlmc&list=PLrjKTqI3jnm TWSXXvWX1 jX-L6f1QJSx&index=39
40.	regular expressions in theory of automata TOC Lec-40 Bhanu Priya	https://www.youtube.com/watch?v=9FG0REWfuL0&list=PLrjKTqI3jnm TWSXXvWX1 jX-L6f1QJSx&index=40
41.	regular expression Examples TOC Lec-41 Bhanu Priya	https://www.youtube.com/watch?v=nNMD1wE3TDM&list=PLrjKTqI3jnm TWSXXvWX1 jX-L6f1QJSx&index=41
42.	convert regular expression to finite automata TOC Lec-42 Bhanu Priya	https://www.youtube.com/watch?v=hiJuCKLQEjQ&list=PLrjKTqI3jnm TWSXXvWX1 jX-L6f1QJSx&index=42
43.	convert regular expression to finite automata examples TOC Lec-43 Bhanu Priya	https://www.youtube.com/watch?v=5_KRbXPCGWg&list=PLrjKTqI3jnm TWSXXvWX1 jX-L6f1QJSx&index=43
44.	Arden's Theorem	https://www.youtube.com/watch?v=Idl_0mPzZjE
45.	Arden's Theorem	https://www.tutorialspoint.com/automata_theory/ardens_theorem.htm
46.	closure properties of regular languages Part-1 TOC Lec-44 Bhanu Priya	https://www.youtube.com/watch?v=vaTHCo6Ns5I&list=PLrjKTqI3jnm TWSXXvWX1 jX-L6f1QJSx&index=44
47.	closure properties of regular languages Part-2 TOC Lec-45 Bhanu Priya	https://www.youtube.com/watch?v=y1wTU4kjZHs&list=PLrjKTqI3jnm TWSXXvWX1 jX-L6f1QJSx&index=45
48.	pumping lemma for RL TOC Lec-46 Bhanu Priya	https://www.youtube.com/watch?v=5M20ka_sULM&list=PLrjKTqI3jnm TWSXXvWX1 jX-L6f1QJSx&index=46
49.	pumping lemma for RL example	https://www.youtube.com/watch?v=fxOmsdW_tjQ&list=PLrjKTqI3jnm TWSXXvWX1 jX-L6f1QJSx&index=47

	TOC Lec-47 Bhanu Priya	nm_TWSXXvWX1_jX-L6f1QJSx&index=47
Module 2		
50.	context free grammer Introduction TOC Lec-48 Bhanu Priya	https://www.youtube.com/watch?v=d9LrKPZsu40&list=PLrjkTql3jnm_TWSXXvWX1_jX-L6f1QJSx&index=48
51.	context free grammar in automata Example-1 TOC Lec-49 Bhanu Priya	https://www.youtube.com/watch?v=nyjB5xW0tQc&list=PLrjkTql3jnm_TWSXXvWX1_jX-L6f1QJSx&index=49
52.	context free grammar in automata Example-2 TOC Lec-50 Bhanu Priya	https://www.youtube.com/watch?v=OtrBEjBSNnA&list=PLrjkTql3jnm_TWSXXvWX1_jX-L6f1QJSx&index=50
53.	context free grammar in automata Example-3 TOC Lec-51 Bhanu Priya	https://www.youtube.com/watch?v=FORd2Sf9pfQ&list=PLrjkTql3jnm_TWSXXvWX1_jX-L6f1QJSx&index=51
54.	derivation in theory of computation TOC Lec-52 Bhanu Priya	https://www.youtube.com/watch?v=B_Gk_xPP8zc&list=PLrjkTql3jnm_TWSXXvWX1_jX-L6f1QJSx&index=52
55.	leftmost and rightmost derivations Example-1 TOC Lec-53 Bhanu Priya	https://www.youtube.com/watch?v=K_aMajzrKF4
56.	leftmost and rightmost derivations Example-2 TOC Lec-54 Bhanu Priya	https://www.youtube.com/watch?v=GF7PYeA2Pwk
57.	leftmost and rightmost derivations Example-3 TOC Lec-55 Bhanu Priya	https://www.youtube.com/watch?v=fE7A0a0jQNc
58.	derivation tree TOC Lec-56 Bhanu Priya	https://www.youtube.com/watch?v=NLw6OoDeuf0
59.	derivation tree example 1 TOC Lec-57 Bhanu Priya	https://www.youtube.com/watch?v=XrHczBql0cE
60.	derivation tree Example 2 TOC Lec-58 Bhanu Priya	https://www.youtube.com/watch?v=lgM3NYDREy4
61.	ambiguity in grammar TOC Lec-59 Bhanu Priya	https://www.youtube.com/watch?v=Mdl2Ti7zefY
62.	ambiguity in grammar example 1 TOC Lec-60 Bhanu Priya	https://www.youtube.com/watch?v=7iFtFJUkw2k
63.	ambiguity in grammar example 2 TOC Lec-61 Bhanu Priya	https://www.youtube.com/watch?v=r5639CLulk0
64.	unambiguous grammar TOC Lec-62 Bhanu Priya	https://www.youtube.com/watch?v=SMEMtRVbx0
65.	unambiguous grammar example 1 TOC Lec-63 Bhanu Priya	https://www.youtube.com/watch?v=BQ2rPI9gp3A
66.	unambiguous grammar example 2 TOC Lec-64 Bhanu Priya	https://www.youtube.com/watch?v=Bbr3u5bNnYY
67.	simplification of CFG TOC Lec-65 Bhanu Priya	https://www.youtube.com/watch?v=603fO3xPIAo

	removal of useless symbols CFG TOC Lec-66 Bhanu Priya	https://www.youtube.com/watch?v=XnteU4oGMuY
69.	elimination of epsilon production TOC Lec-67 Bhanu Priya	https://www.youtube.com/watch?v=zlM_yChkLIM
70.	removal of unit production CFG TOC Lec-68 Bhanu Priya	https://www.youtube.com/watch?v=kidmQlu0lrU
71.	pushdown automata (PDA) in TOC Introduction Part-1/2 Lec-78 Bhanu Priya	https://www.youtube.com/watch?v=DTV9RPlsdW4
72.	pushdown automata (PDA) in TOC Introduction Part-2/2 Lec-79 Bhanu Priya	https://www.youtube.com/watch?v=ANey0U92VBQ
73.	pushdown automata example Part-1/2 TOC Lec-82 Bhanu Priya	https://www.youtube.com/watch?v=ntrF_KxHn18
74.	pushdown automata example Part-2/2 TOC Lec-83 Bhanu Priya	https://www.youtube.com/watch?v=gU3scx0EOJI
75.	Theory of Computation: PDA Example ($a^n b^{2n}$)	https://www.youtube.com/watch?v=K_kYGkeP3L8
76.	Deterministic And Non Deterministic Push Down Automata with Examples Difference PDA	https://www.youtube.com/watch?v=TrwCrPYUDjw
77.	Theory of Computation: Conversion of PDA to CFG	https://www.youtube.com/watch?v=ntMdkSpR6os
78.	cfg to pda conversion conversion TOC Lec-84 Bhanu Priya	https://www.youtube.com/watch?v=mehO_4R7D6o
79.	pumping lemma for CFL TOC Lec-75 Bhanu Priya	https://www.youtube.com/watch?v=ww4m4UJTbWE
80.	pumping lemma example Part-1 TOC Lec-76 Bhanu Priya	https://www.youtube.com/watch?v=91Ukv82n6T4
81.	pumping lemma example Part-2 TOC Lec-77 Bhanu Priya	https://www.youtube.com/watch?v=Dfkv6BDixas
Module 3		
82.	Chomsky normal form TOC Lec-69 Bhanu Priya	https://www.youtube.com/watch?v=uNRMAU8t4I
83.	converting CFG to CNF steps TOC Lec-70 Bhanu Priya	https://www.youtube.com/watch?v=JTontnRjiNc
84.	converting CFG to CNF example TOC Lec-71 Bhanu Priya	https://www.youtube.com/watch?v=7G0PwGrdlH8
85.	griebach normal form TOC Lec-72 Bhanu Priya	https://www.youtube.com/watch?v=G_xUKj1oGL8
86.	converting CFG to GNF example-1 TOC Lec-73 Bhanu Priya	https://www.youtube.com/watch?v=447aBp1M6C4
87.	converting CFG to GNF example - 2	https://www.youtube.com/watch?v=1cEpXxP0z-E

	TOC Lec-74 Bhanu Priya	
88.	Top Down parser Types of Parser Compiler Design Lec - 10 Bhanu priya	https://www.youtube.com/watch?v=a4qdbCQJhyA
89.	Backtracking Parser Top-Down Compiler Design Lec - 11 Bhanu priya	https://www.youtube.com/watch?v=Txr2dg24D_s
90.	Predictive Parser LL(1) Top-Down Compiler Design Lec -12 Bhanu priya	https://www.youtube.com/watch?v=8wAh9zJei4M
91.	LL parser Predictive Parser (Top-down) Compiler Design Lec - 13 Bhanu priya	https://www.youtube.com/watch?v=Kc-uE8yK51A
92.	LL(1) parser First & Follow functions Compiler Design Lec-14 by Bhanu Priya	https://www.youtube.com/watch?v=WfTNV2UEj2M
93.	LL1 parser Example-1 Compiler Design Lec-15 Bhanu Priya	https://www.youtube.com/watch?v=d28MKEBXb6A
94.	LL1 parser Example-2 Compiler Design Lec-16 Bhanu Priya	https://www.youtube.com/watch?v=d2cYkGnBbeM
95.	LR parsing Introduction Compiler Design Lec-17 Bhanu Priya	https://www.youtube.com/watch?v=LIYZ0qSn6-k
96.	LR (0) parsing Items Part-1/3 Compiler Design Lec-18 Bhanu Priya	https://www.youtube.com/watch?v=ZjQst1xnCul
97.	LR (0) parsing Table Part-2/3 Compiler Design Lec-19 Bhanu Priya	https://www.youtube.com/watch?v=eccgl2WWhY4
98.	LR (0) parsing Input string Part-3/3 Compiler Design Lec-20 Bhanu Priya	https://www.youtube.com/watch?v=Qa3IbLnX0DM
99.	SLR(1) parsing example LR(0) items Compiler Design Lec-21 Bhanu Priya	https://www.youtube.com/watch?v=sWBSQoKNDUo
100.	CLR (1) parsing LR(1) items Part-1/3 Compiler Design Lec-22 Bhanu Priya	https://www.youtube.com/watch?v=9BpEbPpYUHs
101.	CLR(1) parsing LR(1) items Part-2/3 Compiler Design Lec-23 Bhanu Priya	https://www.youtube.com/watch?v=IBGM2Fookvg
102.	CLR(1) parsing Table LR(1) items Part-3/3 Compiler Design Lec-24 Bhanu Priya	https://www.youtube.com/watch?v=MoRZTcDkr5A
103.	LALR (1) parsing LR(1) items Compiler Design Lec-25 Bhanu Priya	https://www.youtube.com/watch?v=Nuls6Ut7FQw

	Lexical Analysis using LEX tool Implementation Part-1/2 Compiler Design Lec-27 Bhanu Priya	https://www.youtube.com/watch?v=082oGUHypZU
105.	Lexical Analysis using LEX tool Implementation Part-2/2 Compiler Design Lec-28 Bhanu Priya	https://www.youtube.com/watch?v=9BBdGWiBouE
106.	Parser using YACC tool Implementation Part - 1/2 Compiler Design Lec-29 Bhanu Priya	https://www.youtube.com/watch?v=HoU7deiqs-U
107.	Parser using YACC tool Implementation Part - 2/2 Compiler Design Lec-30 Bhanu Priya	https://www.youtube.com/watch?v=aic-OmXe6yw
Module 4		
108.	Turing machine (TM) Introduction Part-1/2 TOC Lec-85 Bhanu Priya	https://www.youtube.com/watch?v=5xpB9r13fYQ
109.	Turing machine (TM) Introduction Part-2/2 TOC Lec-86 Bhanu Priya	https://www.youtube.com/watch?v=STh3_6E5PQQ
110.	Turing machine(TM) language TOC Lec-89 Bhanu Priya	https://www.youtube.com/watch?v=CRIMCRm0-Ak
111.	turing machine Example-1 TOC Lec-90 Bhanu Priya	https://www.youtube.com/watch?v=uSpeMO52Wxo
112.	turing machine Example-2 TOC Lec-91 Bhanu Priya	https://www.youtube.com/watch?v=EDsmAKJ8_fm
113.	variants of TM Multitape & Non deterministic Part-1/2 TOC Lec-92 Bhanu Priya	https://www.youtube.com/watch?v=Zz2bVMJ1wIU
114.	variants of TM Multihead & Offline Part-2/2 TOC Lec-93 Bhanu Priya	https://www.youtube.com/watch?v=ll_IVVZZOwI
115.	TOC - MODULE 5 - TOPIC 9 - LINEAR BOUNDED AUTOMATA (LBA)	https://www.youtube.com/watch?v=aJInO1bEXA
116.	Abstract Syntax tree (AST) Polish Intermediate Code Compiler Design Lec-37 Bhanu Priya	https://www.youtube.com/watch?v=8_o9T_aJ_zY
117.	Three address Code Intermediate Code Compiler Design Lec-36 Bhanu Priya	https://www.youtube.com/watch?v=0MdPdOmxMUI
118.	Three Address, Quadruples, Triples and Indirect Triples	https://www.youtube.com/watch?v=jNbSX8dID3g
119.	Type checking Compiler Design	https://www.youtube.com/watch?v=TQVAKby6oI

	Lec-39 Bhanu Priya	
120.	Type Expressions & System Compiler design Lec-40 Bhanu Priya	https://www.youtube.com/watch?v=mPhPppowIRs
121.	Type Conversion Implicit & Explicit Compiler design Lec-41 Bhanu Priya	https://www.youtube.com/watch?v=94kgGENcWS0
122.	Symbol table Implementation Compiler Design Lec-42 Bhanu Priya	https://www.youtube.com/watch?v=yJmwYwgqIGU
123.	Code Optimisation technique Compiler Design Lec-51 Bhanu Priya	https://www.youtube.com/watch?v=Cv9ZzALVR5o
124.	Loop Optimisation Code motion & Strength reduction Compiler Design Lec - 52 Bhanu Priya	https://www.youtube.com/watch?v=Yx7_i8wx2sc
Module 5		
125.	2.10 Chomsky hierarchy	https://www.youtube.com/watch?v=azCfOP57fbE
126.	Undecidable and Decidable problem	https://www.youtube.com/live/E5R1NmCzuqc?app=desktop https://www.youtube.com/watch?v=JfX7VK7ocRU
127.	Recursive & Recursively enumerable languages TOC Lec-94 Bhanu Priya	https://www.youtube.com/watch?v=mX9IUltwO0s
128.	halting problem Turing Machine(TM) TOC Lec-95 Bhanu Priya	https://www.youtube.com/watch?v=0Q9qAM2htlI
129.	Code generator Issues in design Part-1/2 Compiler Design Lec - 53 Bhanu Priya	https://www.youtube.com/watch?v=3SjQ9WRmI4M
130.	Code generator Issues in design Part- 2/2 Compiler Design Lec 54 Bhanu Priya	https://www.youtube.com/watch?v=qnNOaNx0ev0
131.	Construct Quadruples,triples,indirect triples for the statement $(a+b)*(c+d)-(a+b+c)$	https://www.youtube.com/watch?v=sv5Qmq8Hjz4
132.	Copy propagation	https://www.youtube.com/watch?v=LBjIDBVPyYY
133.	Loop invariant code motion with example	https://www.youtube.com/watch?v=n5SMmQ532V8
134.	Peephole optimization in compiler design	https://www.youtube.com/watch?v=id4p10d17KU
135.	Directed Acyclic Graph (DAG) Algorithm Compiler Design Lec-56 Bhanu Priya	https://www.youtube.com/watch?v=ZLt3uMa1btU
136.	Directed Acyclic Graph (DAG) Examples Compiler Design Lec-	https://www.youtube.com/watch?v=Lsj_xYQ-TA

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